

Lecture 4

Encryption – Part III (and Pseudo-randomness)

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Adapted from the textbook slides

Outline

- Block ciphers.
- Pseudorandom permutations (PRPs).
- Defining security of encryption.
- Encryption modes.

Block Ciphers

- A pair of algorithms E_k and D_k (encrypt and decrypt with key k) with domain and range of $\{0,1\}^n$
 - Encrypt and decrypt data in blocks each of which is of size n bits.
- Conventional correctness requirement: $m = Dk(Ek(m))$
- Security requirement of block ciphers is to be a pair of Pseudorandom Permutations (PRP).

So what is a Random Permutation?

And what is a PRP?

What is a random permutation ρ ?

- Random permutation ρ over finite domain D , usually: $\{0,1\}^m$
- How can we select a random permutation ρ ?
- Let $D = \{x_1, x_2, \dots, x_n\}$
- For $i = 1, \dots, n$:
 - $\rho(x_i) \overset{\$}{\leftarrow} D - \{\rho(x_1), \rho(x_2), \dots, \rho(x_{i-1})\}$
- Examples:

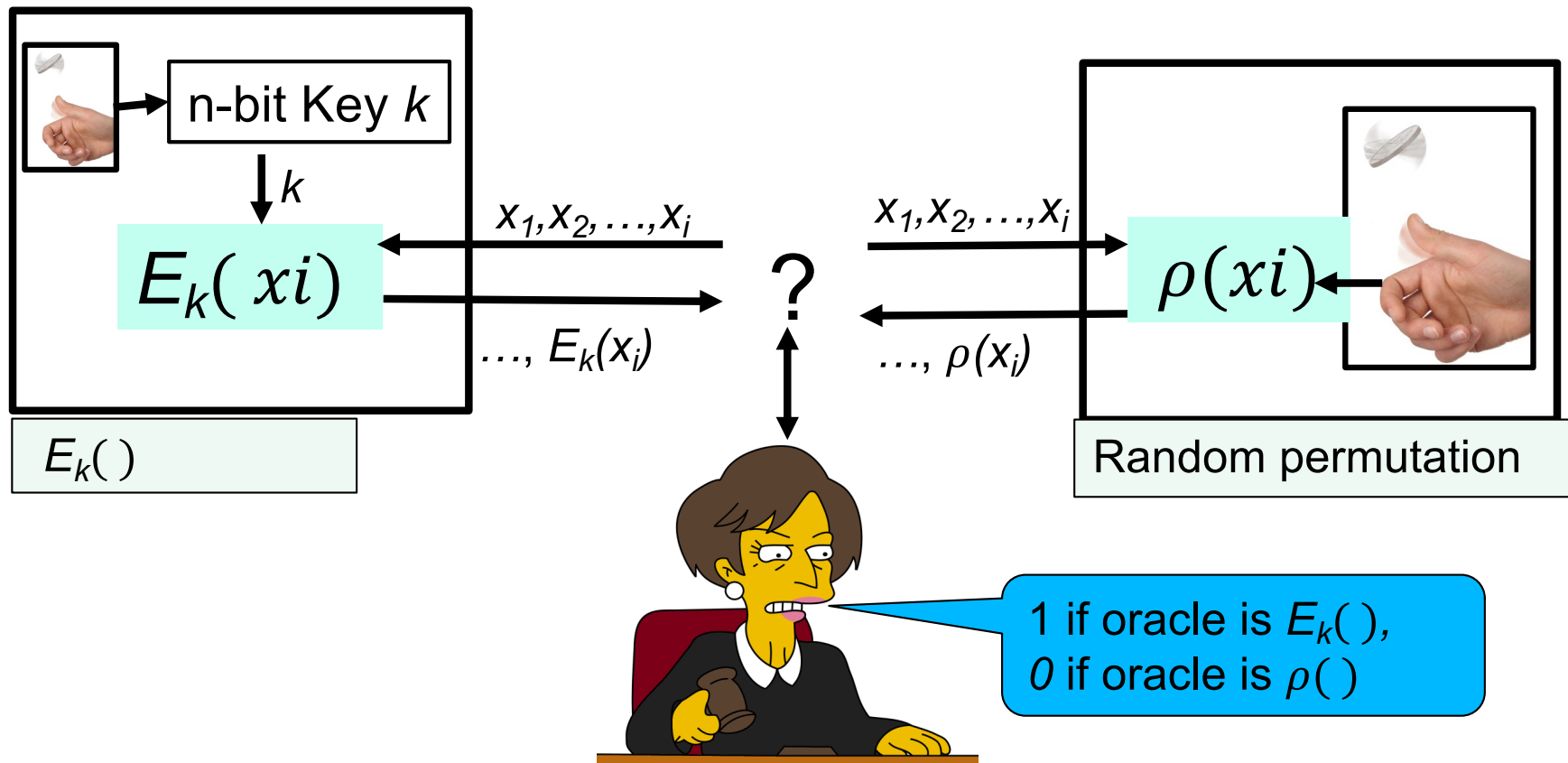
Domain D $\{0,1\}^2$		$\rho()$
	00	10
	01	11
	10	00
	11	01

Domain D $\{0,1\}^2$		$\rho()$
	00	00
	01	01
	10	10
	11	11

Pseudorandom Permutation (PRP)

and their Indistinguishability Test

- E is a PRP over domain D , if no distinguisher or attacker A can distinguish E from a random permutation with non-negligible probability.
 - So A 's advantage is negligible.



$$\epsilon_{\mathcal{A}, E}^{PRP}(n) \equiv \Pr_{k \xleftarrow{\$} \{0,1\}^n} [\mathcal{A}^{E_k}(1^n)] - \Pr_{\rho \xleftarrow{\$} \text{Perm}(D)} [\mathcal{A}^{\rho}(1^n)]$$

Block Cipher: Invertible PRP (E, D)

- A secure block cipher is an invertible PRP.
- Invertible PRP is:
 - A pair of PRPs (E,D) such that $m = D_k(E_k(m))$
 - And (E,D) is indistinguishable from (ρ, ρ^{-1}) so the advantage defined below must be negligible
 - where ρ is a random permutation

$$\varepsilon_{\mathcal{A},(E,D)}(n) \equiv \Pr_{k \xleftarrow{\$} \{0,1\}^n} [\mathcal{A}^{E_k,D_k}(1^n)] - \Pr_{\rho \xleftarrow{\$} \text{Perm}(D)} [\mathcal{A}^{\rho,\rho^{-1}}(1^n)]$$

- Note: A block cipher is deterministic $\rightarrow$ not secure encryption!
 - But it is used to construct encryption (we will study that soon!)

Examples

- ❑ Given $E_k(m) = m + k \bmod 2^n$ where n is a large integer, answer the following:
 - ❑ What is the algorithm D_k ?
 - ❑ Is this construction a secure block cipher? Why?

- ❑ Given $E_k(m) = m \oplus k$, answer the following:
 - ❑ What is the algorithm D_k ?
 - ❑ Is this construction a secure block cipher? Why?

Constructing a PRP

- ❑ Focus: constructions from a PRF $F_k()$
 - ❑ PRFs seem easier to design (less restrictions)
- ❑ The PRF/PRP switching lemma (Lemma 2.2 in the textbook) shows that **every pseudorandom function can also be used as a pseudorandom permutation.**
 - ❑ That is, a PRF over a domain D is indistinguishable from a PRP over D , and vice versa: no efficient adversary is likely to distinguish between a PRP and a PRF (over domain D).

Constructing a block cipher (i.e., an invertible PRP)

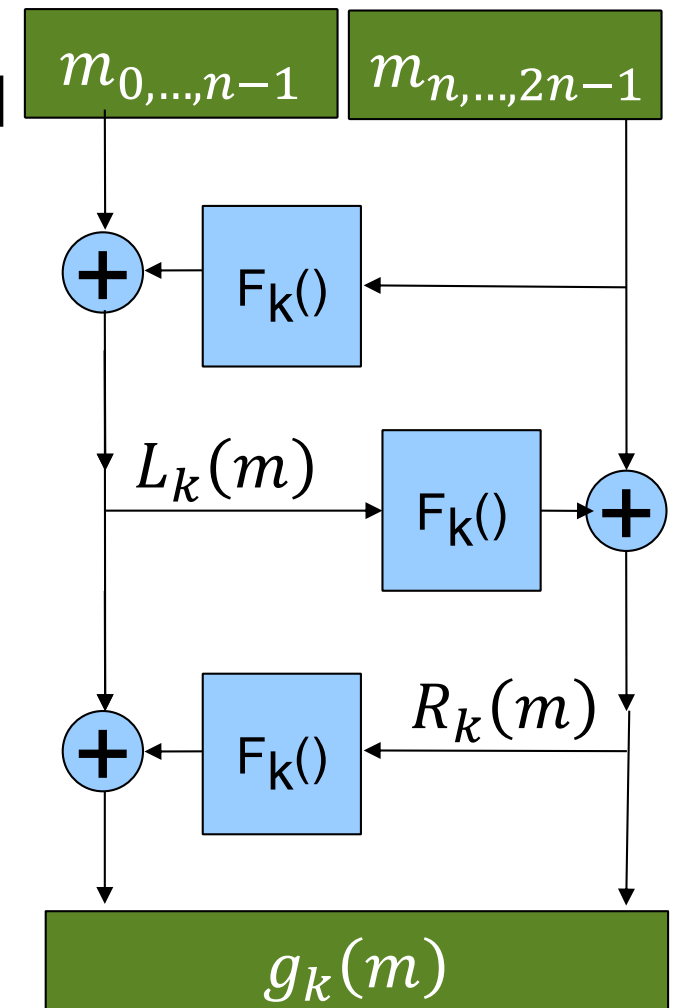
- ❑ A PRF $F_k(\cdot)$ can be used as a PRP
- ❑ **But a block cipher requires an invertible PRP!!!**
- ❑ Challenge: how to make a PRF invertible?
 - ❑ One solution: **The Feistel Construction**

The Feistel Block-cipher Construction

- Turn PRF F_k into a block cipher
 - Three 'rounds' suffice for security [LR88]

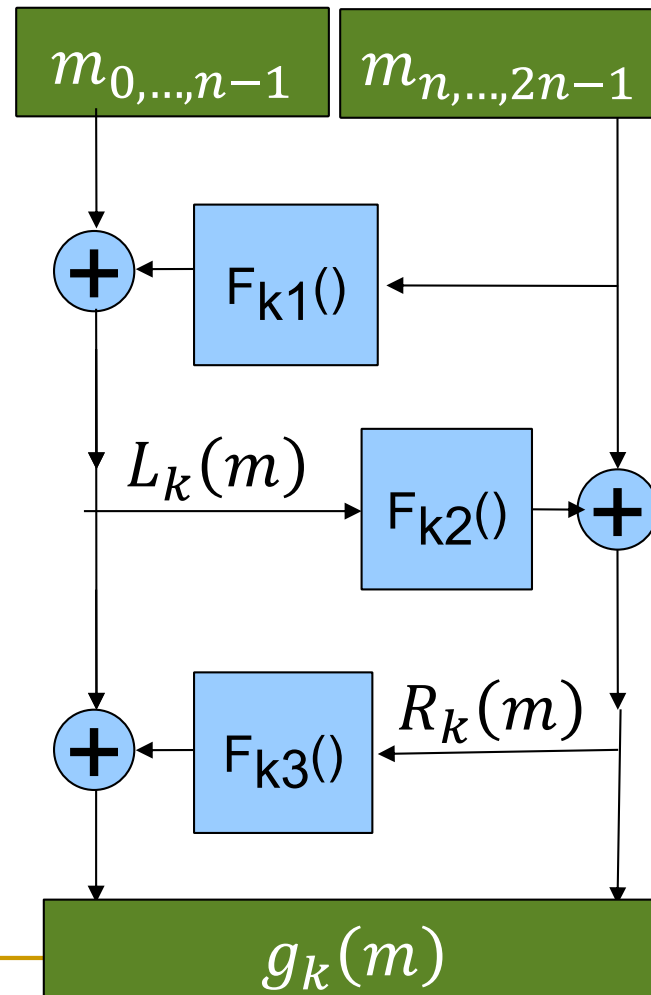
$$\begin{aligned}L_k(m) &= m_{0,\dots,n-1} \oplus F_k(m_{n,\dots,2n-1}) \\R_k(m) &= F_k(L_k(m)) \oplus m_{n,\dots,2n-1} \\g_k(m) &= L_k(m) \oplus F_k(R_k(m)) \oplus R_k(m)\end{aligned}$$

- Used in DES (but not in AES)
 - With 16 'rounds'
 - (DES and AES are symmetric key encryption schemes used in practice, AES replaced DES.)



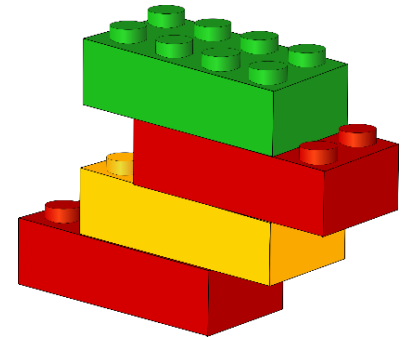
The Feistel Block-cipher Construction

- The secure construction uses different round keys (derived from k). For decryption the order of these keys is reversed.



Cryptography Building Blocks Principle

- Construct schemes from basic functions
 - Provably secure constructions:
attack on scheme → attack on function
 - Allows replacing broken functions
 - Allows upgrading to more secure/efficient functions
- E.g., encryption from block cipher (or PRG/PRF/PRP)
 - Block cipher, PRG, PRF, PRP: **deterministic, stateless, FIL** (Fixed-Input-Length)
 - Encryption: **randomized, can be stateful or stateless, VIL** (Variable-Input-Length)



We defined security for PRG, PRF,
PRP, and Block ciphers.

**How about defining security of
encryption??**

Defining Secure Encryption

- Attacker capabilities:
 - Computational limitations → PPT
 - Ciphertext only (COA), Known / chosen plaintext attack (KPA/CPA), Chosen ciphertext (CCA)?
- What's a successful attack?
 - Key recovery ?
 - May be impossible yet weak encryption scheme...
 - (Full) Message recovery?
 - How about partial exposure, e.g., $m \in \{\text{"Advance"}, \text{"Retreat"}\}$
 - Answer: attacker 'wins' for exposing any info on plaintext

Conservative Design Principle

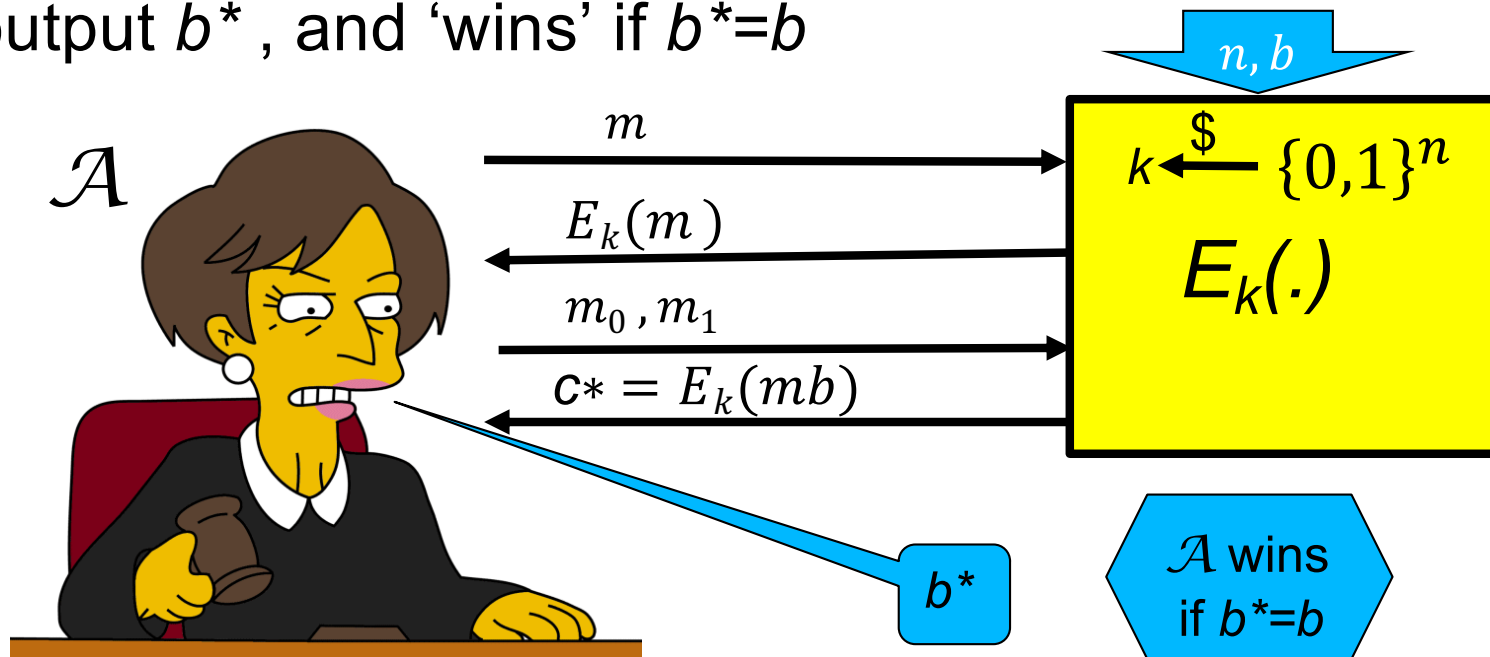
- When designing, evaluating a cryptosystem...
 - Consider most powerful attacker ($\text{COA} < \text{KPA} < \text{CPA} < \text{CCA}$)
 - Be as general as possible – cover many applications
 - And ‘easiest’ attacker-success criteria
 - Not full message/key recovery!
 - Make it easy to use securely, hard to use insecurely!
- When designing, deploying a system (that uses some cryptographic primitives)
 - Restrict attacker’s capabilities (e.g., avoid known/chosen plaintext)

Cryptanalysis Success Criteria for Encryption

- Learn anything at all about plaintext – how to define?
Can we achieve it ?
 - Well-defined notion: ‘semantic security’ [crypto course]
- So an encryption scheme is secure if the attacker cannot learn anything about the plaintext that he did not know in advance.
- **Indistinguishability:** Eve ‘wins’ if she can distinguish between encryptions of (any) two messages
 - The attacker chooses these two messages.
 - We focus on indistinguishability for CPA attacker. In crypto course: equivalent to semantic security

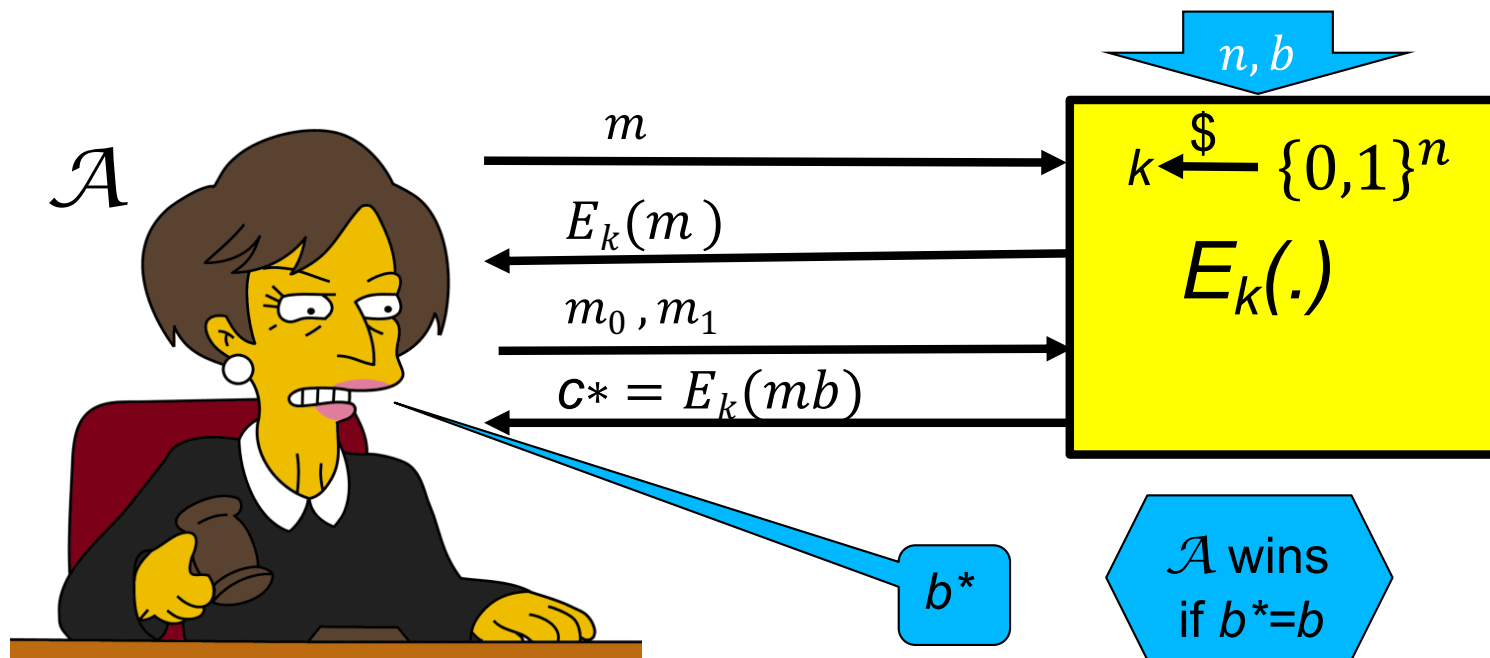
IND-CPA-Encryption Test (1st try)

- ❑ Flip coins to select random bit b and key k
- ❑ $\mathcal{A}$ (adversary) gives message m , receives $E_k(m)$
 - ❑ Can ask for a polynomial number of encryptions (i.e., can make a polynomial number of encryption oracle queries over any message it wishes)
 - ❑ **Chosen Plaintext Attack (CPA)**
- ❑ $\mathcal{A}$ gives two messages (m_0, m_1) , receives $c^* = E_k(m_b)$
- ❑ $\mathcal{A}$ output b^* , and 'wins' if $b^* = b$



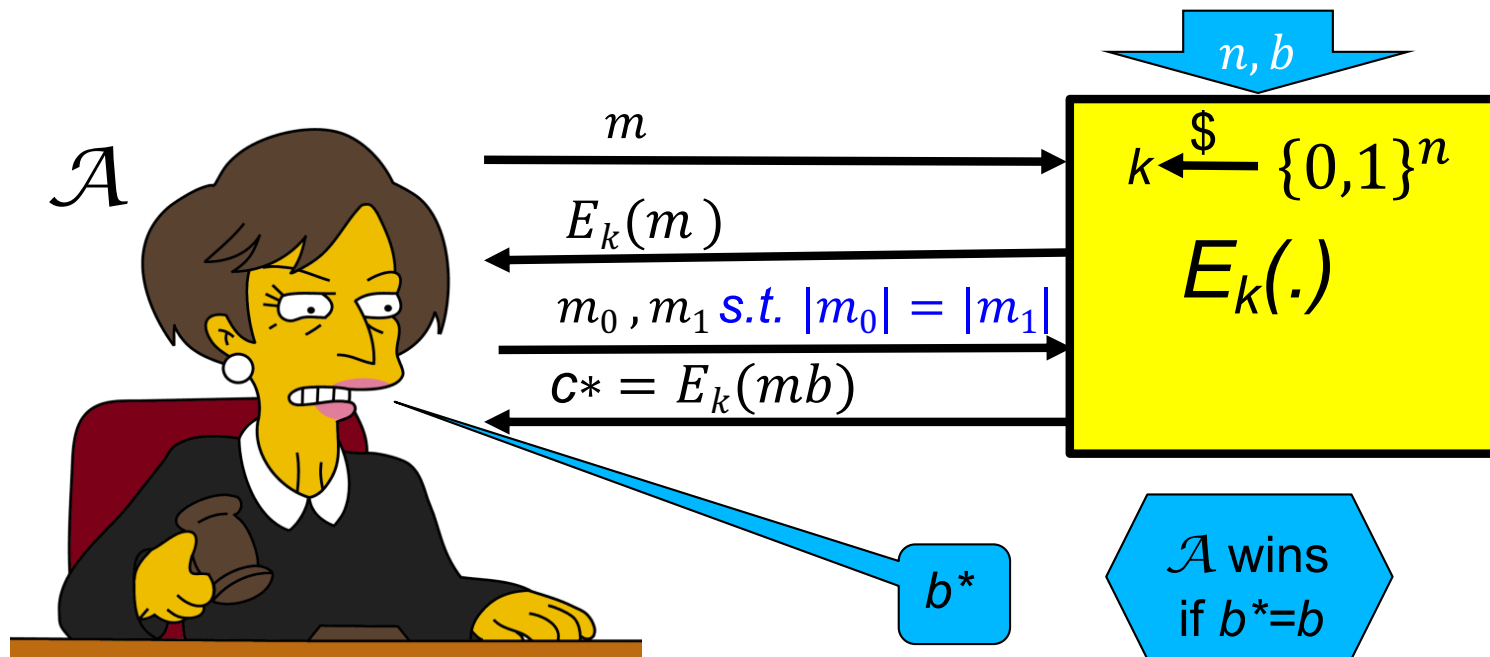
IND-CPA-Encryption Test (1st try): too easy

- ❑ This test is too easy!! The adversary can easily win!!
- ❑ How?
- ❑ Hint: messages can be arbitrary binary strings
 - ❑ Namely, $m, m_0, m_1 \in \{0,1\}^*$
 - ❑ **Solution:** let $m_0=0$, $m_1=11111111111111111111$
 - ❑ If $c^*=E_k(m_b)$ is 'short', output $b^*=0$; if 'long', output $b^*=1$



IND-CPA-Encryption Test (fixed)

- ❑ Flip coins to select random bit b and key k
- ❑ $\mathcal{A}$ (adversary) gives message m , receives $E_k(m)$
 - ❑ Chosen Plaintext Attack
- ❑ $\mathcal{A}$ gives messages (m_0, m_1) s.t. $|m_0| = |m_1|$, receives $E_k(m_b)$
- ❑ $\mathcal{A}$ output b^* , and 'wins' if $b^* = b$



Definition: IND-CPA Encryption

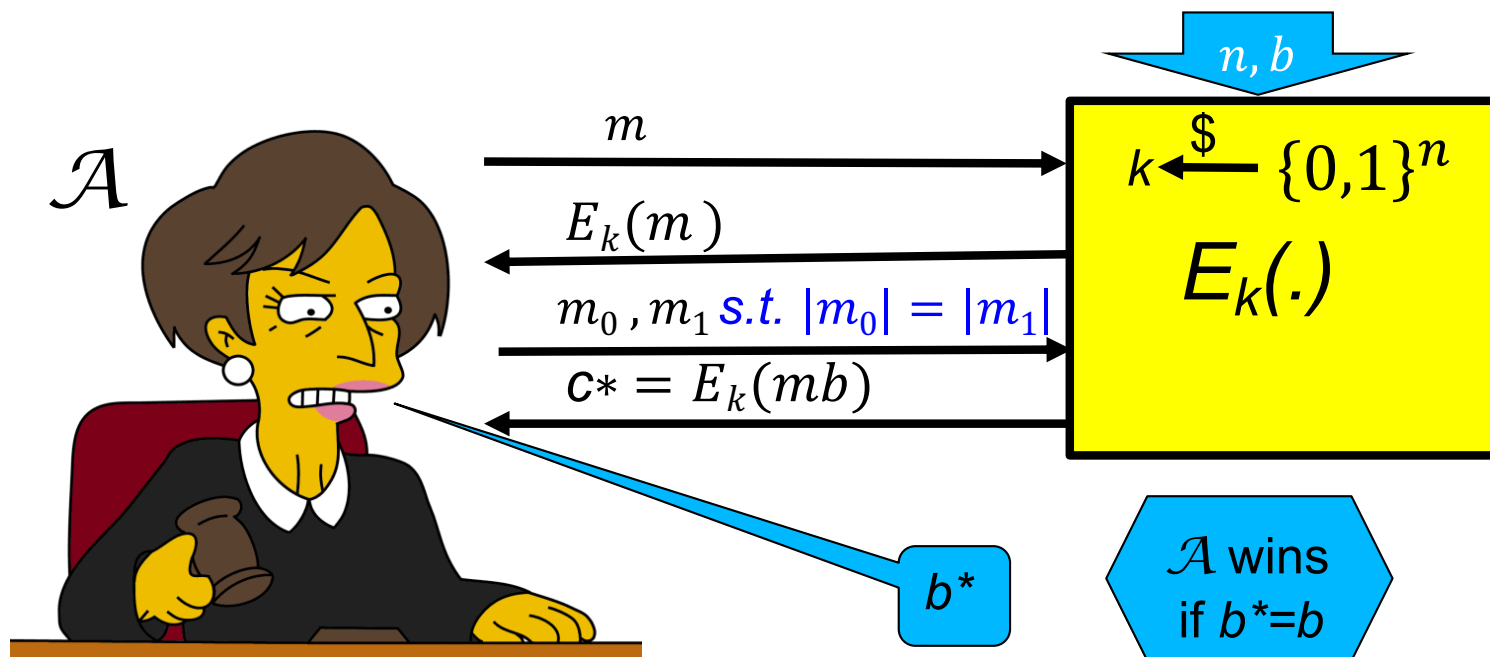
Shared (or symmetric) key encryption (E, D) is **IND-CPA**, if every efficient adversary A has negligible advantage:

$$\varepsilon_{\langle E, D \rangle, \mathcal{A}}^{IND-CPA}(n) \equiv \Pr \left[T_{\mathcal{A}, \langle E, D \rangle}^{IND-CPA}(1, n) = 1 \right] - \Pr \left[T_{\mathcal{A}, \langle E, D \rangle}^{IND-CPA}(0, n) = 1 \right]$$

$$\begin{aligned} T_{\mathcal{A}, \langle E, D \rangle}^{IND-CPA}(b, n) \{ \\ & k \xleftarrow{\$} \{0, 1\}^n \\ & (m_0, m_1) \leftarrow \mathcal{A}^{E_k(\cdot)}(\text{'Choose'}, 1^n) \text{ s.t. } |m_0| = |m_1| \\ & c^* \leftarrow E_k(m_b) \\ & b^* = \mathcal{A}^{E_k(\cdot)}(\text{'Guess'}, c^*) \\ & \text{Return } b^* \\ \} \end{aligned}$$

Can IND-CPA encryption be deterministic?

- ❑ **No!!** But why? Suppose $E_k(m)$ is deterministic...
- ❑ $\mathcal{A}$ can ask E_k to encrypt m_0 and m_1 and then check which one is equal to the challenge ciphertext $\rightarrow$ always wins!
- ❑ **Conclusion: IND-CPA Encryption must be randomized**
 - ❑ *Even if you encrypt the same m over and over again, a new ciphertext will be produced.*



What's next?

Construct a secure encryption scheme?

... provably secure without assumptions ?

Unlikely: Proof of security $\Rightarrow P \neq NP$

(similar argument to PRF)

Instead, let's build secure encryption from PRFs !

(That is, PRF is secure $\Rightarrow$ encryption is IND-CPA)

Actually, we'll use **block cipher** (recall the PRF/PRP switching lemma) to build encryption schemes under what is called “Modes of operation.”

Examples

- ❑ Let F be a secure PRF:
 - ❑ $E_k(m) = F_k(0^n) \oplus m$
 - ❑ *What is the decryption algorithm?*
 - ❑ *Is this construction a secure encryption scheme? Why?*
 - ❑ $E_k(m) = (r, F_k(r) \oplus m)$ where r is a random string freshly generated for each encryption invocation.
 - ❑ *What is the decryption algorithm?*
 - ❑ *Is this construction a secure encryption scheme? Why?*

Encryption: Modes of Operation

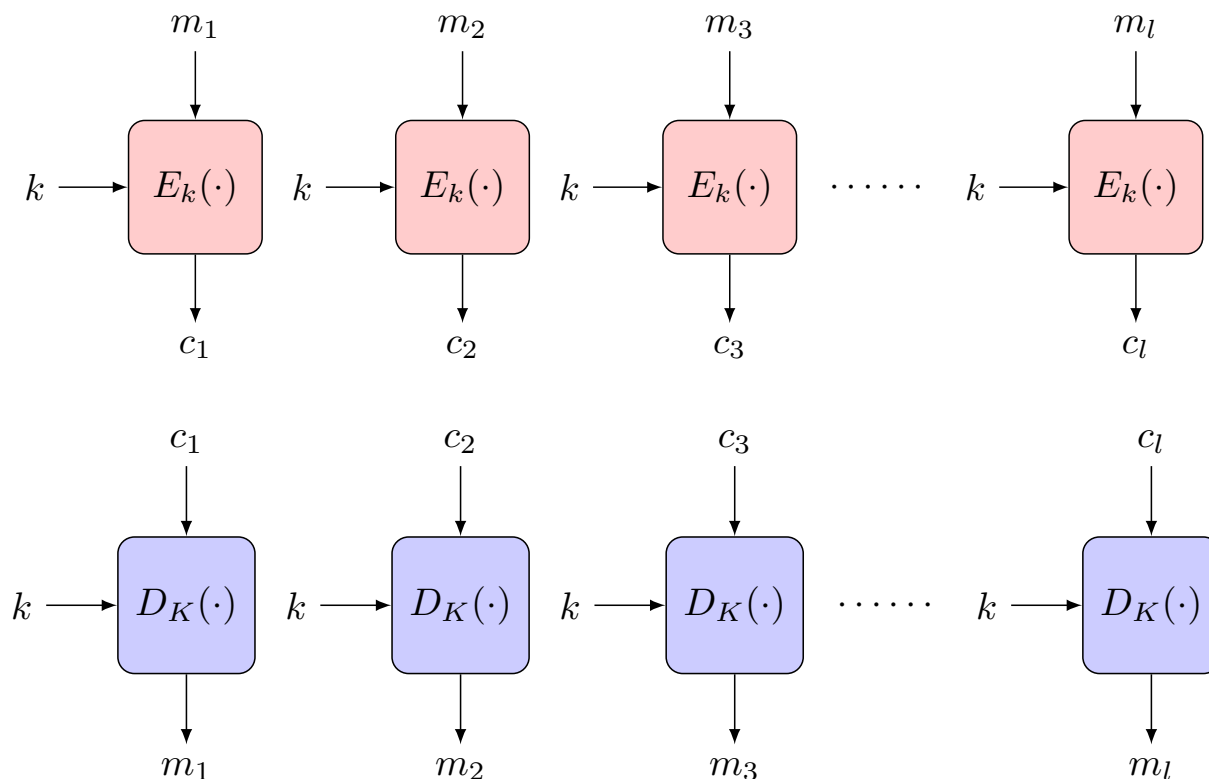
- ❑ 'Modes of operation': use block cipher (PRP), to encrypt long (Variable Input Length, VIL) messages
- ❑ Randomize/add state for security
 - ❑ Often: use random or stateful *Initialization Vector (IV)*
- ❑ Use long keys
 - ❑ Better security (at least against exhaustive search)
- ❑ Assume plaintext message is in blocks: $m_0 || m_1 || \dots$
 - ❑ An integer number of blocks, each block is n bits.

Encryption Modes of Operation

Mode	Encryption	Flip bit $c_i[j] \Rightarrow$	Properties
Electronic code book (ECB)	$c_i = E_k(m_i)$	Corrupt m_i	Insecure (not IND-CPA)
Counter (CTR) [simplified]	$c_i = m_i \oplus E_k(s)$ $s \leftarrow s + 1$ (stateful)	Flip $m_i[j]$	Stateful, parallelizable, preprocessing
Output Feedback (OFB)	$r_0 \xleftarrow{\$} \{0, 1\}^n, r_i = E_k(r_{i-1}),$ $c_0 \leftarrow r_0, c_i \leftarrow r_i \oplus m_i$	Flip $m_i[j]$	Parallelizable, preprocessing
Cipher Block Chaining (CBC)	$c_0 \xleftarrow{\$} \{0, 1\}^n,$ $c_i \leftarrow E_k(m_i \oplus c_{i-1})$	Flip $m_{i+1}[j]$ $\Rightarrow$ corrupt m_i	Parallel decryption

Electronic Code Book mode (ECB)

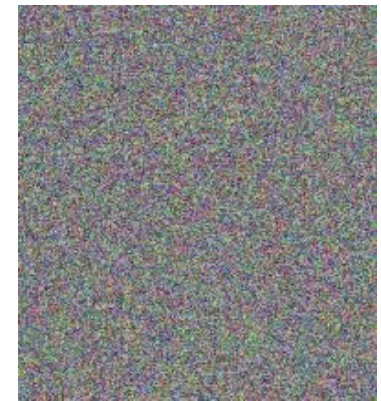
- Encryption $c_i = E_k(m_i)$, decryption $m_i = D_k(c_i)$
 - Each message m is composed of l blocks, $m_1 \dots m_l$, and each m_i is n bit block, and same for c and c_i



Electronic Code Book mode (ECB)

- Encryption $c_i = E_k(m_i)$, decryption $m_i = D_k(c_i)$

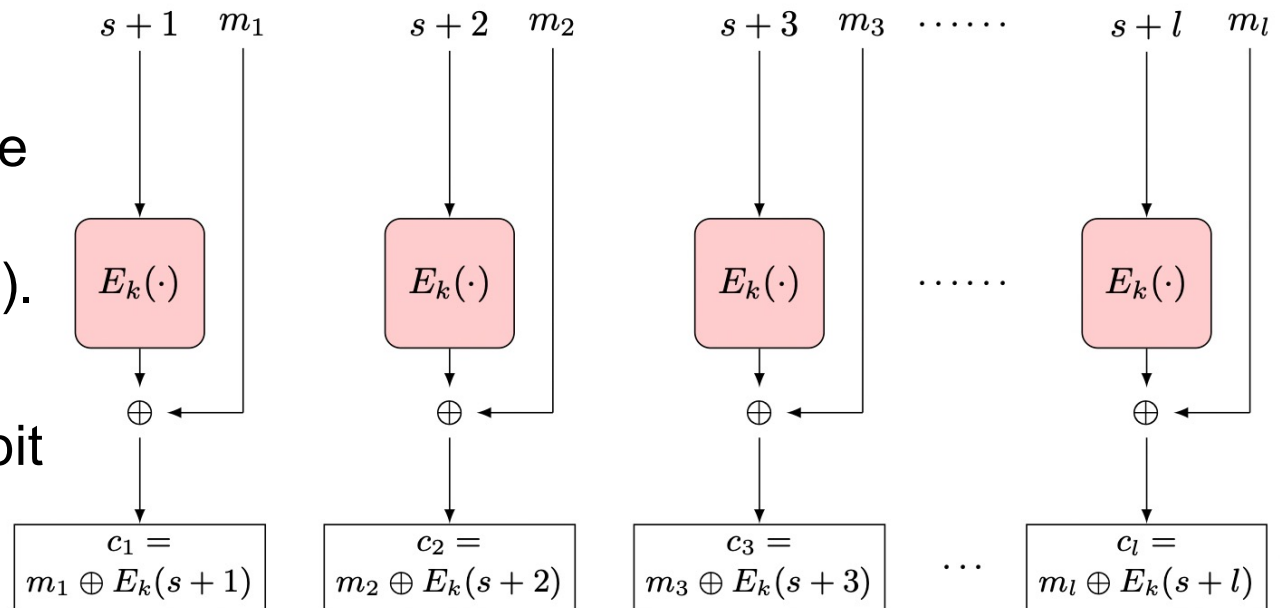
Insecure!! (do not use it!) Which of these is ECB encryption? Why?



Counter (CTR) Mode

- Random counter (or 'initialization vector', IV , or s)
 - $i > 0$: $c_i = E_k(s + i) \oplus m_i$
 - $m_i = E_k(s + i) \oplus c_i$
- Parallel encryption and decryption with offline precomputing

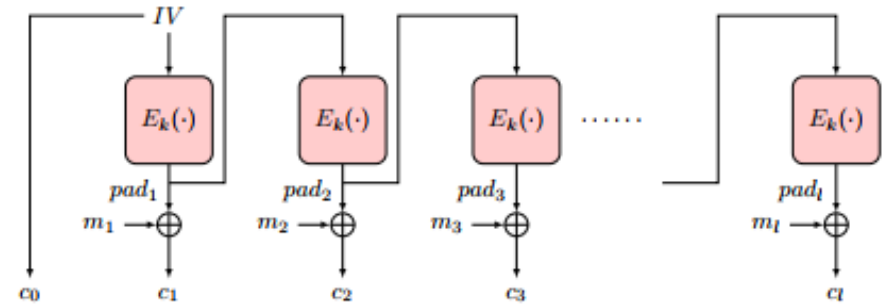
- How to decrypt?
- If a PRF is used as the PRP (or E_k), then it is CPA (provably secure).
- Error propagation:
 - flip bit in $c_i \rightarrow$ flip bit in m_i



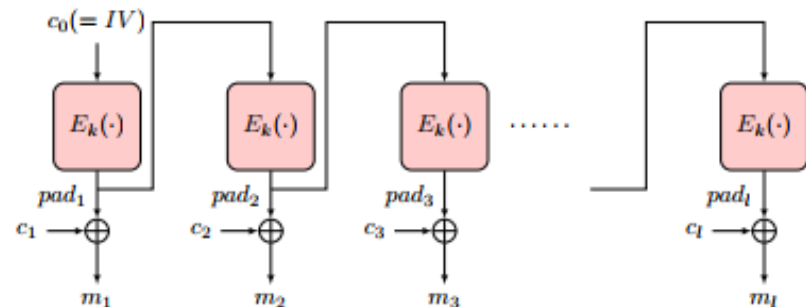
Output-Feedback (OFB) Mode

- Goal: encrypt long (multi-block) messages, with **less random bits**
- How? Use random bits only for first block ('initialization vector')
 - To encrypt next blocks of message, use output of previous block
 - Namely, a **block-by-block stream cipher**

- Encryption: $pad_0 \leftarrow IV$,
 $pad_i \leftarrow E_k(pad_{i-1})$,
 $c_0 \leftarrow pad_0, c_i \leftarrow pad_i \oplus m_i$



- Decryption:
 $pad_0 \leftarrow c_0$,
 $pad_i \leftarrow E_k(pad_{i-1})$,
 $m_i \leftarrow pad_i \oplus c_i$

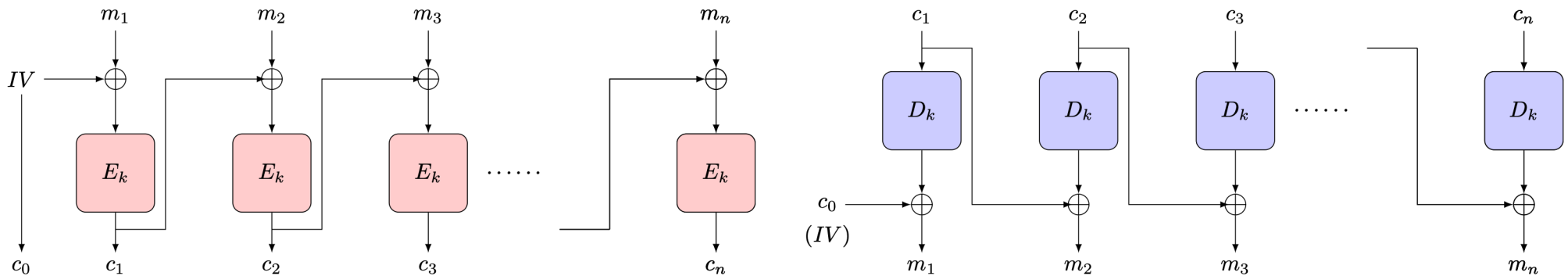


Output-Feedback (OFB) Mode

- **Offline pad computation:** compute pad in advance
 - Online computation: only (parallelizable) XOR !
- Bit errors are bitwise **localized**
 - Corrupting a one bit in the ciphertext corrupts only one bit in the plaintext.

Cipher Block Chaining (CBC) Mode

- Random first block c_0 ('initialization vector', IV)
- $i > 0$: $c_i = E_k(c_{i-1} \oplus m_i)$, $m_i = c_{i-1} \oplus D_k(c_i)$



- Parallel decryption
 - ❑ But no offline precomputing
 - ❑ How about encryption? Sequential (it is a chain!)
- Error propagation:
 - flip bit in $c_i \rightarrow$ flip bit in m_{i+1} and corrupt m_i

Security of CBC mode

- Theorem: If block-cipher E is a (strong) pseudo-random permutation $\rightarrow$ CBC is IND-CPA-secure encryption
- Proof: omitted (crypto course 😊)

Covered Material From the Textbook

- ❑ Sections 2.8, 2.9, and 2.10 excluding:
 - ❑ 2.9.6
 - ❑ 2.10.4
 - ❑ 2.10.7
 - ❑ 2.10.8
 - ❑ 2.10.10
 - ❑ CCA security
 - ❑ Integrity (we will cover this topic separately when we study MACs later)

Thank You!

